

Jonathan Distler

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Education

Cornell University – Ithaca, NY.....May 2027 (Expected)

• Bachelor of Science in Mechanical Engineering with a Physics Minor | GPA: 3.78

• Relevant Coursework:

- Math: Applied Probability and Statistics, Calculus I-III, Differential Equations, Linear Algebra
- Physics: Beginner and Intermediate Fluid Dynamics, Dynamics, Electricity and Magnetism (Honors), Heat Transfer, Mechanics, Mechanics of Engineering Materials, Statics and Mechanics of Solids, Topics in Special Relativity, Thermodynamics
- Application: Lasers and Photonics, Machine Learning for Mechanical Engineers, Mechatronics, System Dynamics

Experience

Valar Atomics Mechanical Engineering Intern

Valar Atomics – Hawthorne, CA.....May 2025 – Present

• Contributed to the successful deployment of the world's **first operational microreactor**, part of a strategic partnership **powering NVIDIA's Blackwell AI infrastructure**

• Designed, detailed, and released **4 production components** (via Solidworks PDM) installed on the **Ward 250 reactor** during criticality; developed Python tools and spreadsheets to model reactor heat transfer losses

• Led **mechanical design** and structural validation of an ASME Sec. VIII Div. I Pressure Vessel

• Created both a **locally hosted website** and an embedded **decision-tree LSTM architecture** to showcase neutronics of the Ward 250 during power ascension and deduce critical streams of data for **predicting criticality**

• Completed **Radiation Worker I & II training**, authorizing entry into controlled radiological areas and higher annual dose limits

Aerospace Adversary Research Intern

Cornell University's Aerospace Adversary Lab – Ithaca, NY.....Jan 2024 – Present

• Collaborating in Dr. Gregory Falco's interdisciplinary aerospace and **cybersecurity research team**, studying autonomous **drone swarms**, and **cyberwarfare** resilience

• Utilizing **Linux**, **ROS 2**, **Gazebo**, **C++**, and **Python** to develop and test **obstacle avoidance** strategies for an autonomous 6-degree-of-freedom drone

• Developed an **Anduril-sponsored** indoor testbed, utilizing GPS with a **Python** and **ROS 2** serial communication system to enable accurate indoor drone control with mapping precision up to **2 cm**

• Developed **Boeing-sponsored** Python simulations to localize adversarial laser sources from irradiance patterns using **eigenvalue decomposition** and **Fisher Information Matrix**-based estimation techniques

• **Co-authored a systems paper on autonomous drone navigation**, submitted for conference publication (*The MAGPIE: Satellite Autonomy for Uncooperative Environments*)

Soft Robotic Fish (SoFi) Research Intern

ETH Zurich's Soft Robotics Laboratory – Zurich, Switzerland.....May 2025 – Aug 2025

• Collaborated in Dr. Robert Katzschman's lab to develop a novel **actuation mechanism** and **autonomous control** for a soft robotic fish tail utilizing a Scotch yoke assembly and a novel radio-frequency methodology under \$80

• Developed a one-to-one **MuJoCo simulation** with **Nelder-Mead optimization**, validating results within 10% of experimental measurements

• Designed and implemented a **motor and IMU control class** to track fish orientation using Euler angles, integrating a Dynamixel motor and Adafruit IMU with sensor fusion and data wrapping techniques.

• Implemented **computer vision** with **remote filming** to measure tail curvature, comparing results with thrust and motor data to optimize tail design and motor actuation rate

Additional Experience

• *Cornell Hyperloop Aeroshell Sub-Team Lead*; Cornell Duffield College of Engineering.....Aug 2026 - Present

• *HRC2 Research Intern*; Cornell Duffield College of Engineering.....Oct 2025 - Present

• *Cornell Hyperloop Structures Sub-Team Member*; Cornell Duffield College of Engineering.....Sept 2024 - Aug 2026

• *Biomechanics Research Intern*; Virginia Tech College of Engineering.....Jun 2024 - Aug 2024

• *Human-Drone Interaction Research Intern*; Virginia Tech (Interdisciplinary)Dec 2023- Jun 2024

• *Stormwater Maintenance Assistance*; Stormwater Solutions LLC.....Aug 2019 - Aug 2023

• *CS Genome Project Research Intern*; Virginia Tech Department of Computer Science.....May 2022 - Sept 2022

Campus Involvement

• *Nuclear is Clean Energy (NiCE) Member*; Cornell University.....Aug 2025 - Present

• *Cornell Brazilian Jiu Jitsu Club Member*; Cornell University.....Jan 2025 - Present